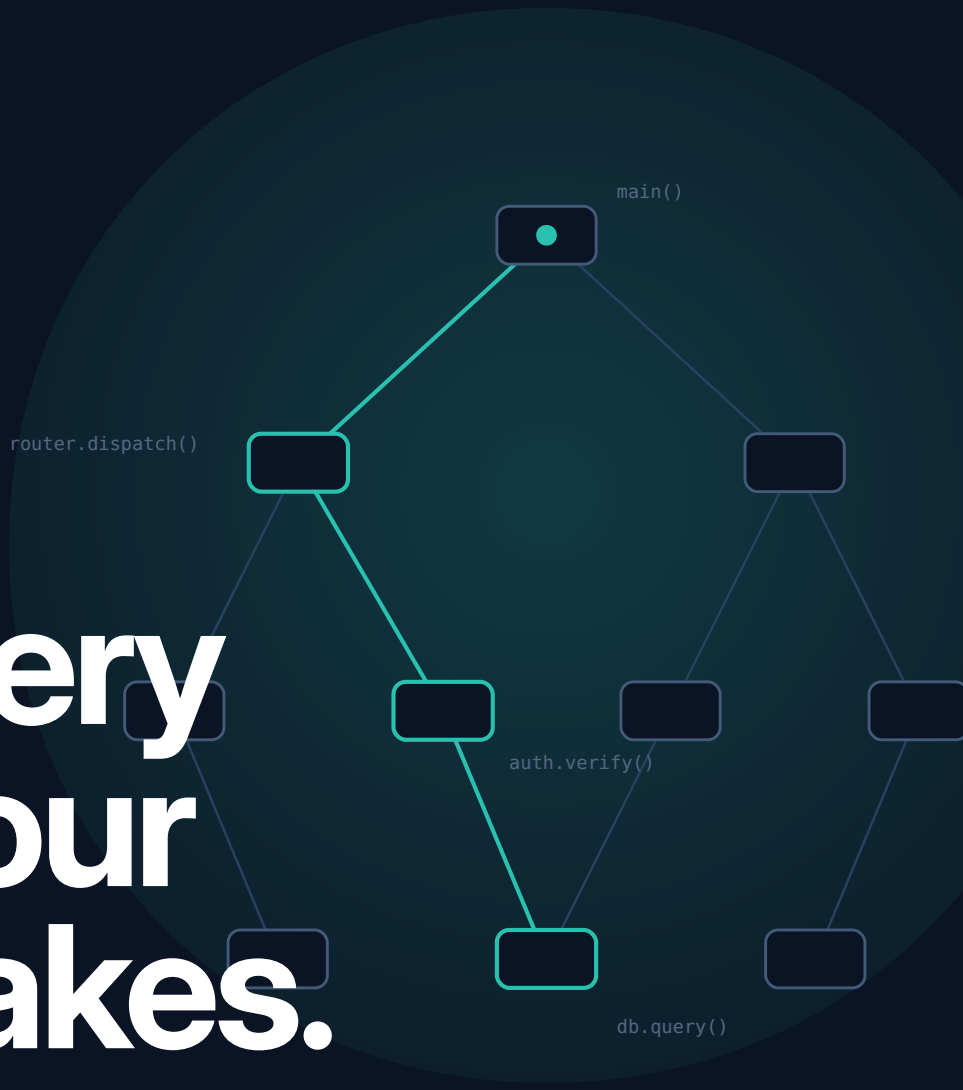
LANTERN · WORKING TITLE



See every path your code takes.

Product specification for a full-stack code-mapping, explanation, and deployment-readiness platform built for AI-assisted developers.

DOCUMENT	DATE	OWNER	STATUS
Product Specification Version 0.4 · Draft	October 31, 2026	Little Henderson	For review · Not for distribution

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Document control

Working title	Lantern. A placeholder; trademark and domain checks are pending.
Scope	Product, architecture, and phased delivery for v1. Pricing and go-to-market are sketched, not decided.
Conventions	"MUST", "SHOULD", and "MAY" mark requirement strength. Performance numbers are targets to validate in Phase 0, not measured results.

Revision 0.4	Adds chapter 20: Lantern as a connector inside AI assistants (one MCP server for Claude, ChatGPT, Cursor and Claude Code), the "Send to…" handoff, the badge wording, the Connections screen, and the three plugin seams (analysis packs, explanation providers, exporters). All of it is implemented and tested in the repository.
Revision 0.3	Adds the competitive landscape (chapter 18) and a prioritized improvement roadmap mapped to the app's screens (chapter 19). DeepSeek becomes the default model, with a free hosted allowance followed by bring-your-own-key (section 8.5). The allowance gateway is implemented and tested in the repository.
Revision 0.2	Full stack becomes first-class: TypeScript/JavaScript moves into Phase 1 alongside Python, the API bridge (section 6.5) and full-stack security checks (9.5) are added, and Rust and C++ move to later phases. A working prototype of all three exists in the repository.
Third-party tools	Named tools reflect the landscape as of October 2026. Licenses and capabilities change, so each choice gets re-checked before build.

01 · EXECUTIVE SUMMARY

Executive summary

Lantern maps every call path in a full-stack app, from the button a user clicks to the database query it causes, explains each piece in three registers, audits it with real security tooling, and grades how ready it is to ship. The output is evidence a builder can hand to a skeptical engineer.

A new class of builders ships software with AI assistance. They move fast, but they often can't explain what their code does, and they don't know what they don't know about security or deployment. Professional engineers notice, and the label "AI slop" sticks even when the work is sound.

Lantern closes that gap in four moves:

1. **Map.** It parses the frontend (TypeScript/JavaScript, including React and Next.js) and the backend (Python first; Rust and C++ later) into a single call graph, links each frontend API call to the backend route that serves it, and renders the result as an interactive flowchart starting from real entry points: pages, API routes, CLI commands, `main()`.
2. **Explain.** Every function and module gets three explanations: a technical one, a plain-English one with analogies, and a mapped one where each everyday phrase is pinned to the exact technical term it stands for.
3. **Audit.** Established static-analysis and dependency scanners find the security problems. The call graph tells you which findings are actually reachable. An LLM of the user's choosing explains each finding and drafts a fix, and a rescan verifies the fix.
4. **Prove.** A Deployment Readiness Score and a shareable Evidence Report tie each claim to a specific commit and to the tool that produced it.

TWO WAYS IN

Cloud mode

Connect GitHub. A sandboxed worker clones the repo, analyzes it, and destroys the copy. Zero setup, works from any device.

TWO WAYS IN

Local agent mode

Run one binary on your own machine. Source never leaves it; only results are uploaded, with optional redaction. Model calls can stay local too.

WHY FULL STACK, AND WHY NOW

Most apps this audience builds are full stack, and much of that code is TypeScript. The flaws engineers point at often sit where the layers meet: secrets shipped to the browser, API routes the frontend calls without a login check, database access rules left off. A backend-only map can't see those. On the open-source FastAPI full-stack template, a Python-only analysis covers 2,901 of about 13,700 lines; the rest is the React frontend.

THE CORE DESIGN BET

Deterministic tools find facts; language models explain them. The call graph, the security findings, and the readiness checks come from parsers and scanners whose output is repeatable. Models never invent structure or vulnerabilities. They translate verified facts into language people can learn from. That separation is what makes the output credible to engineers.

v1 scope at a glance

AREA	IN V1	LATER
Languages	Python and TypeScript/JavaScript with the frontend-to-backend bridge (Phase 1), Rust (Phase 2), C++ (Phase 3)	Go, Java, Kotlin, Swift
Access	GitHub cloud mode; local agent on macOS, Linux, Windows	GitLab, Bitbucket, self-hosted cloud workers
Models	Bring-your-own-key for major providers, plus local models	Hosted credits, fine-tuned explainer model
Client	Installable PWA with offline report viewing	IDE extensions, native wrappers
Collaboration	Single user, shareable read-only reports	Teams, comments, review workflows

02 · PROBLEM AND AUDIENCE

Problem, audience, and positioning

AI tools let people build software faster than they can understand it. Lantern sells understanding, and the proof of it.

2.1 The gap

An AI-assisted builder can produce a working app in a weekend. What they usually can't produce is an answer to the questions a reviewer asks: *What calls this? What happens if this input is malicious? Where are the secrets? How does this run in production? What breaks under load?* The criticism they get is often fair on the specifics, but it lands as dismissal of the whole person. Nothing on the market today turns "I built this" into "I understand this, and here's the evidence."

2.2 Who it's for

PRIMARY

The AI-assisted builder

Ships real projects with AI coding tools. Self-taught or career-switching. Wants to learn what their code does and stop getting dismissed. Uses a phone and laptop interchangeably.

Job to be done: "Help me understand my code well enough to defend it, and fix what's indefensible."

SECONDARY

The reviewing engineer

Receives an Evidence Report from a builder, a contractor, or a junior teammate. Has five minutes. Trusts scanners more than claims.

Job to be done: "Show me, quickly and verifiably, whether this is safe to merge or deploy."

SECONDARY

The learner and educator

Students, bootcamps, and self-learners who use the Three-Lens view to climb from plain English to real terminology on codebases they care about.

FUTURE

The small team lead

Onboards new hires onto an unfamiliar codebase and wants a living map plus a readiness gate in CI.

2.3 Landscape

CATEGORY	WHAT IT DOES WELL	WHAT'S MISSING FOR THIS AUDIENCE
Call-graph tools	Accurate structure (code2flow, pyan, IDE call hierarchies)	No explanation, no security, no shareable output; developer-only UX
AI repo explainers	Fluent summaries of a whole repository	Structure is inferred by the model, so it can be wrong; no plain-to-technical mapping
Security scanners	Trusted findings	Output assumes security expertise; little guidance on reachability or fixes for beginners
AI code review bots	PR-level feedback	Diff-focused, not whole-system understanding; no learning path

POSITIONING STATEMENT

For AI-assisted builders who get dismissed by engineers, Lantern is the code-understanding platform that turns a repository into a verified map, plain-English explanations, and a security and readiness report. Unlike AI explainers, every structural and security claim comes from deterministic tools and is traceable to a commit.

03 · PRINCIPLES

Product principles

These principles settle design arguments. When two good ideas conflict, the higher one wins.

PRINCIPLE	WHAT IT MEANS IN PRACTICE
1 · Facts from tools, words from models	Call edges, findings, and readiness checks MUST come from deterministic analysis. Models MUST only explain, triage, and propose fixes, and their output is always labeled as model-generated.
2 · The user decides where code goes	Cloud mode and local mode are equal first-class paths. Every screen that sends data anywhere says exactly what is sent and to whom.
3 · Show confidence, never fake certainty	Every edge and finding carries a confidence level. Dynamic languages produce uncertain graphs; Lantern says so instead of hiding it.
4 · Teach, don't just tell	Plain language is a ramp toward real terminology, never a replacement for it. The mapped lens exists so learners graduate.
5 · Evidence must be verifiable	Reports are pinned to a commit hash, list the tools and versions used, and state what they do not prove.
6 · Practice what we preach	Lantern handles source code and API keys. It MUST pass its own audit and publish its security posture.
7 · Mobile is a real surface	Every core task, short of initial local-agent setup, works on a phone.

> **TONE GUARDRAIL**
> Lantern is on the builder's side but never flatters. A low readiness score is shown plainly, with the path to improve it. The report's credibility with engineers depends on it being honest with builders.

04 · FEATURES

Feature overview

Six capabilities, each built on the one before it.

F1

Code Atlas

An interactive map of every function, component, and route, and who calls whom, across frontend and backend. A path can start at a page, pass through a click handler and an HTTP request, and end at a database call. Two views: a zoomable network overview and a top-down Flow view.

F2

Three-Lens explanations

Technical, Everyday, and Mapped explanations for every node, toggled instantly. The Mapped lens pairs each plain phrase with its technical twin, one-to-one.

F3

Security audit

Real scanners, reachability analysis over the call graph, model-written explanations in all three lenses, suggested fixes as diffs, and verification by rescan. Also proactive suggestions for security features the app lacks.

F4

Deployment readiness

A 0–100 score across ten categories, plus a systems-design advisor that recommends a deployment shape and generates starter files such as a Dockerfile, a CI workflow, and `.env.example`.

F5

Evidence Report

A shareable, commit-pinned report with a public link and PDF export, built for an engineer with five minutes.

F6

Model choice

Bring your own API key for any major provider, or run local models. Pick models per task, preview costs before a run, and compare two models' explanations side by side.

4.1 Core user journey

1. **Connect.** The user signs in on the PWA and chooses cloud mode (authorize the GitHub App on selected repos) or local mode (install the agent and pair it by scanning a QR code or entering a one-time code).

2. **Configure.** They choose models for each task and see an estimated token cost for a full run. They choose a privacy level (Section 5.3).
3. **Analyze.** Structural analysis and scanning run first and finish quickly. Explanations stream in afterward, entry points first. A push notification fires when the run completes.
4. **Explore.** They open the Atlas at an entry point, follow the path, and toggle lenses on any node.
5. **Fix.** They work down the findings and readiness checklist, apply suggested patches (cloud mode opens a pull request; local mode writes a patch file or applies it on confirmation), then re-run.
6. **Prove.** They publish an Evidence Report and share the link.

05 · ARCHITECTURE

System architecture and the two access modes

One analysis engine, packaged as a single binary, runs in two places. That guarantees identical results whether the user picks cloud or local.

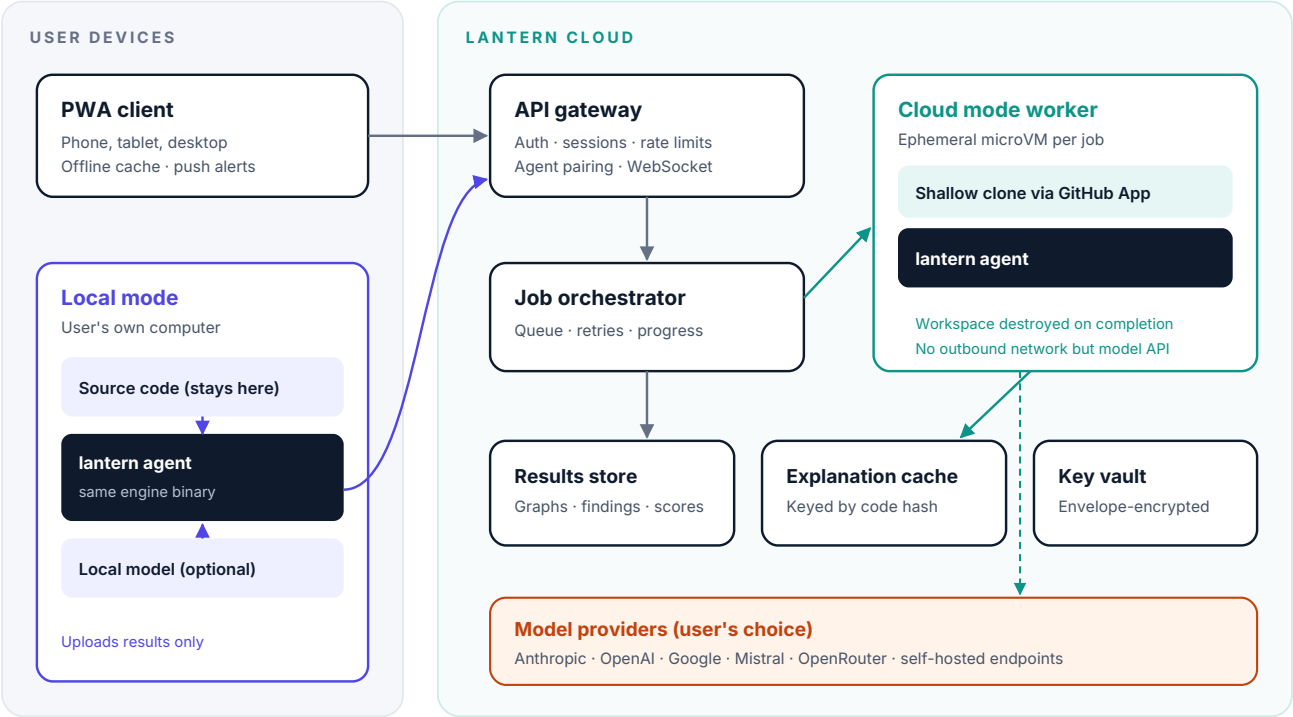


Figure 1. System overview. The same `lantern agent` binary performs analysis in both modes. In cloud mode it runs inside a disposable sandbox; in local mode it runs on the user's machine and sends only results.

5.1 Cloud mode

Best for zero setup and for users who work mostly from a phone or tablet.

- **Access.** A GitHub App with the narrowest permissions: read-only repository contents and metadata on repositories the user explicitly selects. Pull-request write permission is requested separately, only when the user first asks Lantern to open a fix PR.
- **Isolation.** Each job runs in a fresh microVM (Firecracker or gVisor class) with no persistent disk, CPU and memory caps, a wall-clock limit, and egress restricted to the configured model endpoints.
- **Untrusted code.** The worker never runs the user's build scripts or tests by default. This matters most for C++ (CMake configure steps execute code) and Rust (build scripts and procedural macros execute code). Section 6.3 covers the accuracy trade-off and the opt-in "sandboxed build" setting.
- **Retention.** The clone is destroyed when the job ends. Only derived results persist: the graph, findings, scores, and explanations. Source snippets inside results are governed by the privacy level.

5.2 Local agent mode

Best for private or proprietary code and for users who want model calls to stay on their own hardware.

- **Distribution.** A single signed binary for macOS (Apple silicon and Intel), Linux (x86-64 and ARM64), and Windows. Install via Homebrew, a shell script with a published checksum, winget, or `cargo install`.
- **Pairing.** `lantern login` shows a one-time code and QR code. The user approves it in the PWA, and the agent receives a device-scoped, revocable token.
- **Operation.** `lantern analyze` runs once. `lantern watch` re-analyzes on file changes. The agent can also sit idle and accept "analyze now" requests triggered from the PWA, so a user on their phone can kick off a run on their home machine.
- **Model calls.** Made directly from the agent using keys stored in the operating system's keychain, or sent to a local model server. In this mode Lantern's servers never see API keys or source.
- **Tooling.** The agent uses language servers and scanners already installed on the machine when present, and offers to fetch pinned versions when they're missing.

5.3 What leaves the machine: privacy levels

Both modes honor one setting that controls how much source-derived content is stored in Lantern's cloud. The PWA shows the active level on every analysis.

LEVEL	STORED IN LANTERN CLOUD	TRADE-OFF
Full	Graph, findings, explanations, and short source snippets (the function body shown beside its explanation)	Best experience; code viewable from any device
Structure only	Graph, symbol names, file paths, findings, explanations. No source text.	Code panel only works when the agent is online
Redacted	As above, with names and paths replaced by stable aliases. The alias map stays on the user's machine.	Shared reports show aliases unless the user publishes the map
Offline	Nothing. The agent serves the PWA over the local network or localhost.	No remote access or sharing

> **BE PRECISE WITH USERS**
> Even in local mode, if the user chooses a cloud model provider, source code is sent to that provider to generate explanations. The UI MUST say this in plain words when a cloud model is selected, and MUST offer local models as the alternative. "Your code never leaves your computer" is only true with a local model.

5.4 Mode comparison

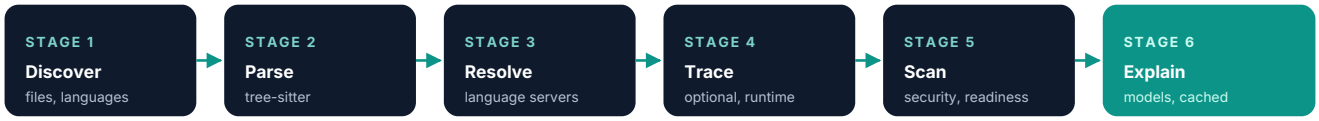
CONSIDERATION	CLOUD MODE	LOCAL AGENT MODE
Setup	Two taps: sign in, select repos	Install binary, pair once
Where source lives	Briefly in a disposable sandbox	Never leaves the machine (except to a chosen cloud model)
Analysis accuracy	Good; build-dependent analysis limited by default	Best; can use the user's real build environment
Runtime tracing	Opt-in, test suite only, in sandbox	Full: run the app or tests and record real calls
Private or non-GitHub repos	GitHub only in v1	Any folder, any host
API key custody	Encrypted in Lantern's vault	OS keychain; never sent to Lantern
Applying fixes	Opens a pull request	Writes a patch, or applies on confirmation

06 · ANALYSIS ENGINE

Analysis engine

The engine turns a repository into a Unified Code Graph: one language-neutral model of every symbol and every call, with an honest confidence level on each edge.

6.1 Pipeline



Stages 1–5 are deterministic and run in seconds to minutes. Stage 6 is the only stage that calls a model, and it streams results as they arrive.

Figure 2. Analysis pipeline. Each stage writes to the Unified Code Graph; later stages enrich it rather than replace it.

6.2 Unified Code Graph

All languages are normalized into one graph schema, so the PWA, the scanners' reachability step, and the explainer never need language-specific logic.

NODE KIND	EXAMPLES
Module	Python module, Rust crate or module, C++ translation unit
Type	Class, struct, trait, enum, interface
Callable	Function, method, closure, lambda
EntryPoint	`main`, route handler, CLI command, test
External	Third-party library call, system call, network or database boundary

EDGE KIND	MEANING
calls	A invokes B
may_call	A invokes one of a set (dynamic dispatch)
contains	Module or type owns a member
implements	Type fulfills a trait or base class
reads / writes	Touches a file, environment variable, socket, or database

Edge confidence

Every edge carries one of four confidence levels, displayed in the Atlas as line style so users can see at a glance what is proven and what is inferred.

LEVEL	LINE STYLE	SOURCE
Observed	Solid, bold	Recorded during a real run (Stage 4)
Resolved	Solid	A language server or compiler-grade resolver confirmed the target

LEVEL	LINE STYLE	SOURCE
Inferred	Dashed	Name and type heuristics; plausible but unconfirmed
Possible	Dotted	One of several candidate targets of dynamic dispatch

6.3 Language support

LANGUAGE	PARSE AND RESOLVE	HARD CASES	RUNTIME TRACING
Python	tree-sitter for syntax; Pyright (or an equivalent type-aware analyzer) for resolution. Import graph built from the project's real layout and virtual environment when available.	Monkey-patching, `getattr`, decorators that wrap callables, framework magic (Django signals, FastAPI dependency injection). Framework adapters recognize common patterns.	PEP 669 `sys.monitoring` on Python 3.12+, with a `sys.setprofile` fallback for older versions. Low overhead, records caller→callee pairs only.
TypeScript / JavaScript	tree-sitter (TS, TSX, JS) for syntax; module resolution through relative paths, `tsconfig` path aliases, and barrel re-exports. Later: the TypeScript compiler API for type-aware resolution.	React components (render edges, event handlers, callbacks passed as props), hooks returning objects, framework conventions: Next.js app/pages routers and route handlers, TanStack and React Router, Express/Hono/Fastify. Generated API clients.	Not in v1. Planned through the V8 inspector or a build-time transform so frontend edges get the same precision measurement Python has.
Rust	tree-sitter plus rust-analyzer for name and trait resolution.	Trait objects (`dyn Trait`) become *may_call* sets. Macros are expanded where rust-analyzer can. In cloud mode build scripts and procedural macros are disabled by default because they execute arbitrary code, which lowers accuracy on macro-heavy crates.	Opt-in instrumentation through the `tracing` ecosystem or a sampling profiler; local mode only in v1.
C++	tree-sitter for a fast first pass; clangd driven by `compile_commands.json` for accurate resolution. The agent can generate the compile database from CMake, Meson, or Bear.	Templates, macros, virtual calls, function pointers, and build-flag-dependent code. Without a compile database, accuracy drops sharply; the UI says so prominently.	Opt-in compiler instrumentation (`-finstrument-functions`) or uprobes-based tracers on Linux; local mode only in v1.

> **WHY THE ORDER IS PYTHON + TYPESCRIPT → RUST → C++**
> Python and TypeScript cover most of what the target audience ships, usually together: a React or Next.js frontend on a Python or Node backend. Shipping them together is what makes the frontend-to-backend path visible. Rust has excellent tooling and moderate difficulty. C++ is the hardest to analyze correctly and the riskiest to analyze safely in the cloud, so it ships last.

6.4 Entry-point detection and scale

Large repositories produce graphs too dense to read. Lantern never opens on the whole graph. It detects entry points and starts there:

- **Detected automatically:** `if __name__ == "__main__"`, `fn main`, `int main`, console-script and binary targets in `pyproject.toml`, `Cargo.toml`, and CMake; route decorators for Flask, FastAPI, Django, Axum, and Actix; CLI frameworks such as Click, Typer, argparse, and clap; frontend pages (Next.js app and pages routers, TanStack and React Router), Next.js route handlers, Express-style routes, and SPA mounts; and test functions.
- **Collapsing:** modules collapse into single nodes until expanded. Calls into the standard library and third-party packages collapse into an "External" boundary node by default.
- **Incremental runs:** each callable is fingerprinted by a hash of its normalized syntax tree. Re-runs only re-resolve changed files and only re-explain changed callables, which keeps repeat runs fast and cheap.

6.5 The API bridge: frontend to backend

The bridge turns two half-maps into one. Every HTTP call found in frontend code (`fetch`, axios, generated OpenAPI clients) is matched to the backend route that serves it, and becomes an edge in the graph with its own confidence level.

MATCH	RULE	CONFIDENCE
Exact	Same HTTP method and the same path segments, with any dynamic segment (`{id}`, `:id`, `[id]`, `${id}`) matching any other	Confirmed
Prefix added elsewhere	The route matches the end of the call's path; the backend mounts it under a global prefix such as `/api/v1`	Likely
Unknown base URL	The call starts with a variable (`${API_URL}/items`) and matches the end of a route	Likely
Tie	Several routes match equally well; shape-for-shape matches win first (a variable path segment prefers `/users/{id}` over `/users/me`)	Possible, one edge each
No route	Shown as an external HTTP node: a third-party service, or a backend that isn't in the repository	Stated as unmatched

Prototype results: on the open-source FastAPI full-stack template, all 23 frontend calls to its API link to the correct Python route, verified mechanically against the generated client's method names. On the Next.js app *taxonomy*, all 5 calls to its own API link exactly, checked by hand against every `fetch` in the repository.

07 · EXPLANATIONS

The Three-Lens explanation system

Each node can be read three ways. The third lens is the product's signature: it teaches the vocabulary instead of hiding it.

7.1 The three lenses

LENS	AUDIENCE AND RULES
1 · Technical	For engineers. Precise terminology, inputs and outputs, side effects, error behavior, complexity where relevant, and the role the function plays in its call path.
2 · Everyday	For anyone. No jargon. Must include at least one concrete example ("if a user types…") and one analogy. Analogies come from a single *metaphor world* chosen per repository, so explanations stay consistent across the whole map.
3 · Mapped	A bridge. The everyday explanation, rewritten so that every plain phrase is paired 1:1 with the exact technical term, symbol, or line it stands for. Tapping either side highlights its twin.

7.2 Worked example

Function: `auth.verify_token(token: str) -> User` in a FastAPI app. Metaphor world for this repository: *a hotel*.

LENS 1 · TECHNICAL

Decodes a JWT with HS256 using the `JWT_SECRET` environment variable, validates `exp` and `iss` claims, then loads the user by the `sub` claim through `UserRepo.get()`. Raises `HTTPException(401)` on any decode or lookup failure. Called by the `get_current_user` dependency on every protected route.

LENS 2 · EVERYDAY

This is the front desk checking a room key. When someone tries to enter a members-only part of the app, they show a key card. The desk checks that the card was made by this hotel, hasn't expired, and belongs to a real guest. If anything is off, the guest is turned away. For example, a card issued yesterday that was only good for one hour gets rejected.

LENS 3 · MAPPED

the front desk	↔ `verify_token()`
a key card	↔ `token: str (a JWT)`
made by this hotel	↔ `signature check, HS256 + JWT_SECRET`
hasn't expired	↔ `exp claim validation`
belongs to a real guest	↔ `UserRepo.get(sub)`
turned away	↔ `raise HTTPException(401)`
members-only part of the app	↔ `routes using get_current_user`

7.3 Keeping the mapping honest

A plain-English layer is only valuable if it's anchored to reality. Lantern enforces that mechanically:

- **Structured output.** Models return a JSON object, not free text (schema below). Each Lens 3 pair carries an `anchor`: a symbol ID from the graph, a source line range, or a vocabulary term from a curated glossary.
- **Anchor validation.** Every anchor is checked against the graph and the source. Pairs with anchors that don't exist are rejected, and the explanation is regenerated or flagged. A model cannot invent a function that isn't there.
- **Grounding context.** The prompt includes the callable's source, its callers and callees from the graph, any security findings on it, and the repository's metadata. Models explain what the graph shows, not what they guess.
- **Glossary build-up.** Technical terms that recur across a repository (JWT, middleware, mutex) gain a glossary entry in all three lenses, so a learner can browse vocabulary on its own.

```
// Explanation record (abridged)
{
  "node_id": "py:auth.verify_token",
  "code_hash": "b3f1…9a",  "model": "provider/model-name",  "prompt_version": "lens-v3",
  "lens1": { "summary": "…", "inputs": [ … ], "side_effects": [ … ], "errors": [ … ] },
  "lens2": { "text": "…", "example": "…", "analogy": "…", "metaphor_world": "hotel" },
  "lens3": [ { "plain": "the front desk", "technical": "verify_token()",
               "anchor": { "type": "symbol", "id": "py:auth.verify_token" } }, … ],
  "validated": true
}
```

7.4 Generation strategy

- **Bottom-up, then top-down.** Leaf functions are explained first, so callers can reference what their callees do. Module and entry-point summaries are generated last from their members' explanations.
- **Lazy by default.** Entry points and their first three levels of calls are explained eagerly. Everything else is explained when first opened, which keeps the first run's cost low. "Explain everything" is one tap, with a cost estimate first.

- **Path narratives.** In Flow view, the user can ask for a step-by-step story of one entry point's path ("what happens when someone logs in"), in any lens.

---

08 · MODELS

Model choice: the multi-LLM layer

Users choose the model for each job, see what it will cost before it runs, and can always tell which model wrote what.

8.1 Provider support

- **Default: DeepSeek** (`deepseek-flash`), free up to an allowance through the Lantern gateway, then on the user's own DeepSeek key (section 8.5).
- **Bring your own key (BYOK)** for major providers: DeepSeek, Anthropic, OpenAI, Google, Mistral, and aggregators such as OpenRouter that expose many models behind one key.
- **Local and self-hosted models** through any OpenAI-compatible endpoint, which covers Ollama, llama.cpp's server, vLLM, and LM Studio.
- **Adapter design.** A thin internal interface with three adapters (OpenAI-compatible, Anthropic, Google) covers nearly every provider. The cloud backend MAY use a gateway library such as LiteLLM for routing, spend tracking, and fallbacks; the agent ships its own small adapters so it stays a single dependency-free binary.
- **Capability checks.** Before a model is selectable for a task, Lantern runs a short probe for structured JSON output and context length. Models that fail the Lens 3 schema reliably are marked "not recommended for Mapped lens."

8.2 Per-task routing

TASK	WHAT IT NEEDS	TYPICAL CHOICE
Lens 1–3 explanations	High volume, structured output, good writing	A fast, inexpensive model
Path narratives and module summaries	Long context, synthesis	A mid-tier model
Security triage and fix drafting	Careful reasoning, low error tolerance	The strongest model available
Systems-design advisor	Broad knowledge, judgment	The strongest model available

Users can accept a recommended routing preset (Economy, Balanced, Best) or set each task by hand.

8.3 Cost, caching, and transparency

- **Cost preview.** Before any model run, Lantern estimates tokens from the graph's size and shows a price range per provider. Users can set a hard spend cap per run and per month.
- **Cache key.** `(code_hash, context_hash, model, prompt_version, lens)`. An unchanged function is never paid for twice. Changing models doesn't discard the cache; both versions are kept.
- **Attribution.** Every explanation, triage note, and fix shows the model that produced it and the date. Reports list all models used.
- **Compare mode.** Any node can be explained by two models side by side. This doubles as a quality check and helps users pick a default.

8.4 Key custody

MODE	WHERE KEYS LIVE
Local agent	The OS keychain (macOS Keychain, Windows Credential Manager, Secret Service on Linux). Never transmitted to Lantern.
Cloud	Envelope-encrypted: a per-user data key encrypts the API key; a managed KMS key encrypts the data key. Decrypted only inside the worker for the job's duration. Never logged, never returned to any client after entry; the UI shows only the last four characters.

8.5 Default model and the free allowance

New users shouldn't need an API key to see Lantern work. The default model is DeepSeek's `deepseek-flash`: it's OpenAI-compatible, supports the JSON output the explainer needs, and is cheap enough to give away. Each user gets a free allowance on Lantern's key. When it's used up, Lantern asks for their own key and carries on where it stopped.

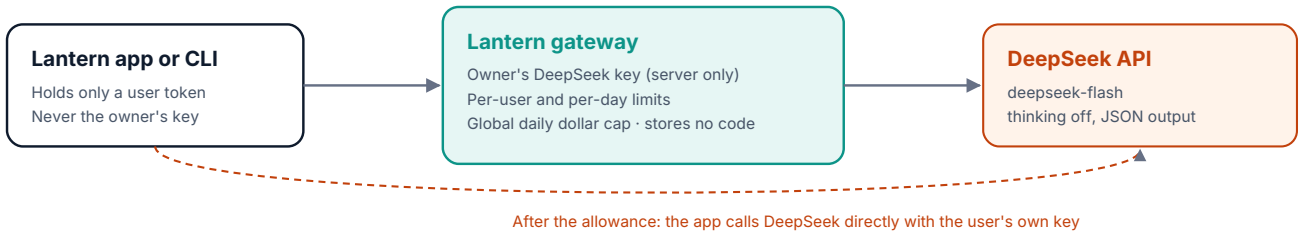


Figure 4. The free allowance. The owner's key lives only on the gateway; clients carry a revocable Lantern token.

Why a gateway is required

Lantern's client code is open source. Any key shipped inside the app or CLI can be extracted within minutes and spent by anyone. The owner's DeepSeek key therefore lives only on a small server, the gateway, which forwards each request and counts it. The app holds a Lantern token that can be revoked at any time and is worthless anywhere else.

What the gateway enforces

CONTROL	DEFAULT	WHY
Free explanations per user	300 requests and 1.5M tokens, whichever comes first	Enough for several full runs on a typical app
Per user per day	120 requests	Stops a single token from draining the budget in an hour
Global daily budget	$10/day at DeepSeek's peak rates; the allowance pauses for everyone until 00:00 UTC	A hard ceiling on the owner's bill, whatever happens
Request shape	One model, `max_tokens` ≤ 3,000, body ≤ 250 KB, plain chat messages only	The gateway can't be used as a general free LLM
Storage	Token hashes, counts, and costs only. No request or response bodies.	The gateway never holds anyone's code

What it costs

An explanation uses roughly 700 input and 1,100 output tokens. At `deepseek-flash` peak prices ($0.30 per million input tokens on a cache miss, $1.20 per million output) that is about **$0.0015**, or $0.003 if the validator triggers a retry. Off-peak prices are half that.

ITEM	ESTIMATE (PEAK PRICES)
One user's full free allowance (300 explanations)	$0.45 typical, $0.90 with retries
1,000 users each using all of it	$450–900, spread over time and capped at $10/day by default
A user on their own key, per 1,000 explanations	About $1–2 paid to DeepSeek

Prices from DeepSeek's API documentation, October 2026. They change; the gateway's spend cap reads its per-token prices from configuration.

The user's journey

1. **Start:** no key needed. The app shows "Free explanations: 300 left" in Settings and next to the Explain button.
2. **First explanation:** a one-time notice says the source of each explained function goes to DeepSeek, quotes DeepSeek's stated data location, and offers a local model instead. Nothing is sent until the user agrees.
3. **Running low (under 10%):** a gentle banner with an "Add your own key" button.
4. **Used up:** the run stops cleanly. Finished explanations are saved, and the user is shown how to add a DeepSeek key, choose another provider, or switch to a local model. Re-running with a key bills only what's left.

DATA LOCATION: SAY IT PLAINLY

DeepSeek's privacy policy says it stores personal data in the People's Republic of China and may use inputs to improve its models. Lantern sends function source code to the model, and its secondary audience (reviewing engineers) is sensitive to exactly this. Three safeguards keep the default credible: the consent notice above, a one-click switch to a local model, and an option to serve the same open DeepSeek models from a host in another jurisdiction if the owner later chooses one. The gateway design doesn't change if the upstream host does.

Screen states for the app (Settings, Explain, errors)

STATE	WHAT THE USER SEES
Settings › Model, default	"DeepSeek (default) · Free explanations: 287 of 300 left · resets never · Add your own key". Simple mode shows only this line; Advanced shows provider, model, and per-task routing.
Settings › Model, own key added	"DeepSeek · Your key (ending 4f2a) · No Lantern limit". A "Remove key" link. The key is stored in the OS keychain, never on Lantern's servers.
Settings › Model, local	"Ollama · llama3.1 on this computer · Your code stays here".
Consent (first explain per provider)	"To write explanations, Lantern sends each function's code to DeepSeek. DeepSeek says it stores data in China and may use it to improve its models. [Continue] [Use a local model instead]".
Error: allowance used up	"You've used your 300 free explanations. Your 41 finished explanations are saved. [Add a DeepSeek key, about $1–2 per 1,000] [Use another provider] [Use a local model]".
Error: daily limit	"You've reached today's free limit. It resets at midnight UTC, or add your own key to keep going now."
Error: allowance paused	"Free explanations are paused for today. Add your own key to continue, or try again tomorrow."
Error: key rejected	"DeepSeek didn't accept this key. Check it at platform.deepseek.com, or switch back to the free allowance."

09 · SECURITY

Security audit

Scanners find; the call graph prioritizes; models explain and draft fixes; a rescan verifies. Each step is labeled so an engineer can see where every claim came from.

9.1 Scanner stack

CONCERN	PYTHON	RUST	C++
Code patterns and taint	Opengrep or Semgrep CE engine; Bandit	Opengrep or Semgrep CE engine; Clippy security lints	Opengrep or Semgrep CE engine; cppcheck; clang-tidy (cert-*, bugprone-*)
Vulnerable dependencies	pip-audit / OSV-Scanner	cargo-audit (RustSec) / OSV-Scanner	OSV-Scanner on Conan lockfiles; inventory of vendored libraries
Unsafe or risky code	Use of `eval`, `pickle`, `subprocess` with shell	cargo-geiger (count of `unsafe`)	Raw buffers, unchecked casts, banned functions list
Secrets	gitleaks across the working tree, including frontend files, and, in local mode, optionally across git history		
Supply chain	Lockfile presence, pinned versions, typosquat heuristics, license inventory		
TypeScript / JavaScript	Opengrep rules for JS/TS; npm audit or OSV-Scanner on `package-lock.json` / `pnpm-lock.yaml`; Lantern's own JS rules for `eval`, raw HTML (`dangerouslySetInnerHTML`, `v-html`), shell and SQL built from template strings, and disabled TLS checks		

LICENSING: REQUIRES LEGAL REVIEW BEFORE LAUNCH

In December 2024 Semgrep moved its maintained rules to the Semgrep Rules License v1.0, which limits them to internal, non-competing, non-SaaS use. The engine itself remains LGPL 2.1. Lantern's cloud mode is a SaaS, so it MUST NOT bundle Semgrep-maintained rules there. The plan: run the Opengrep fork (or the LGPL engine) with Lantern's own rules plus permissively licensed community rules. In local mode, the agent MAY run a user's own Semgrep installation and rules, since that is the user's internal use; counsel should confirm this reading.

9.2 The finding lifecycle

1. **Collect and normalize.** All scanner output is converted to SARIF, deduplicated, and attached to graph nodes by file and line.
2. **Reachability.** Lantern checks whether the affected function is reachable from any entry point, and whether untrusted input (a route parameter, file upload, CLI argument) can flow there along graph edges. Reachable findings rank up; unreachable ones rank down but are never hidden. This is the feature engineers will value most: it separates real risk from noise.

3. **Triage.** The chosen model explains the finding in all three lenses and gives a likelihood that it's a false positive, with reasoning. Models MAY flag a likely false positive; they MUST NOT dismiss one. Only the user can mark a finding as accepted risk, with a written reason that appears in the report.
4. **Fix.** The model drafts a minimal patch as a unified diff, with a Lens 2 explanation of what changed and why.
5. **Verify.** After the patch is applied, the same scanner re-runs on the affected files. A finding is only marked "fixed" when the scanner agrees.

9.3 Severity model

RANK	RULE
Critical	High scanner severity, reachable from an entry point, and touched by untrusted input. Also any live secret.
High	High severity and reachable, or a known-exploited dependency vulnerability.
Medium	Medium severity and reachable, or high severity but unreachable.
Low	Everything else, including hardening suggestions.

9.4 Security features to add

Beyond finding bugs, Lantern recommends protections the app is missing, based on what the graph shows it does. Each recommendation names the evidence that triggered it.

IF THE GRAPH SHOWS…	LANTERN SUGGESTS
Public HTTP routes	Authentication on non-public routes, rate limiting, request size limits, security headers, CORS configured narrowly
Route input reaching a database call	Parameterized queries or an ORM, input validation schemas (Pydantic, serde with validation)
Passwords handled	A slow password hash (Argon2id or bcrypt), never a plain or fast hash
File uploads	Type and size checks, storage outside the web root, randomized file names
Secrets read from code or config files	Environment variables or a secrets manager, plus a committed `.env.example`
Outbound calls built from user input	An allow-list to prevent server-side request forgery
C++ or `unsafe` Rust handling external input	Bounds-checked containers, sanitizer builds (ASan, UBSan) in CI, fuzzing harness for parsers

9.5 Full-stack checks

These rules target the mistakes that cluster in AI-built web apps. All are deterministic: file conventions, patterns, and queries over the graph. Rules that can misfire are labeled as heuristics in the report.

CHECK	WHAT IT LOOKS FOR	RANK
Secrets sent to the browser	Secret-looking environment variables with a browser-exposed prefix (`NEXT_PUBLIC_`, `VITE_`, `REACT_APP_`, SvelteKit public env), read in code or set in `.env` files	High
Database master key in code	A committed JWT whose role claim is Supabase's `service_role`, which bypasses all database rules	Critical
Database open to everyone	Firebase rules with `if true` or test-mode expiry; Supabase tables created without row-level security, or policies `using (true)`	Critical / High
Missing login check (heuristic)	A route that changes data, that the frontend actually calls (via the bridge), with nothing named like an auth check within three calls	Medium
Wildcard CORS	Any-origin CORS in FastAPI, Express, or response headers; high when combined with credentials	High / Medium

Every finding, frontend or backend, also carries a **path from the app**: the page, the event (such as "on click"), and the HTTP request that lead to it. A shell-injection bug in a Python handler is reported as reachable from a specific button on a specific page.

10 · READINESS

Deployment readiness and systems design

A score that can be gamed is worthless. So most checks are deterministic, every point has a visible reason, and the advisor produces real files rather than advice alone.

10.1 The Deployment Readiness Score

Scored 0–100 across ten weighted categories. Weights below are a starting proposal to validate with reviewing engineers in Phase 1.

CATEGORY	WEIGHT	EXAMPLE CHECKS
Security findings	20	No open Critical or High findings; accepted risks documented
Secrets and config	12	No secrets in code; config from environment; `.env.example` present
Input and error handling	10	Validation at boundaries; no bare `except:`; no `unwrap()` on external input; errors don't leak stack traces to users
Dependencies	10	Lockfile committed; no known-vulnerable versions; versions pinned
Tests	10	Tests exist; tests touch every entry point; tests pass (local mode, or opt-in sandbox)
Observability	8	Structured logging; a health-check endpoint for services; no `print` debugging on hot paths
Packaging	8	Reproducible build; container definition with a non-root user and a slim base image
CI/CD	8	A CI workflow that builds, tests, and scans on every push
Resilience	8	Timeouts on outbound calls; retries with backoff; graceful shutdown
Documentation	6	README with run instructions; SECURITY.md; license

10.2 Systems-design advisor

The advisor classifies the project's shape from the graph (web API, background worker, CLI tool, library, desktop app, data pipeline), then recommends a deployment pattern and explains it in all three lenses.

- **Shape-specific advice.** A web API gets twelve-factor configuration, a container, a managed database, a reverse proxy, and horizontal scaling notes. Long-running request handlers get a recommendation to move work onto a queue. A CLI gets release packaging and cross-platform builds.

- **Hotspot detection.** The graph reveals fan-in hotspots (one function everything depends on), synchronous network calls inside loops, and database calls inside request loops (the N+1 pattern). Each is flagged with the path that shows it.
- **Generated starter files.** Dockerfile, `.dockerignore`, a GitHub Actions workflow that runs tests and Lantern's scanners, `.env.example`, SECURITY.md, and a health-check handler. Each is delivered as a diff the user reviews; nothing is committed silently.
- **Architecture diagram.** A one-page diagram of the recommended deployment, which can be included in the Evidence Report.

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11 · EVIDENCE

The Evidence Report

The deliverable that changes the conversation with a skeptical engineer: short, verifiable, and honest about its limits.

11.1 Contents

1. **Header.** Repository, commit hash, branch, analysis date, mode (cloud or local), and privacy level.
2. **Scorecard.** Readiness score with category breakdown; finding counts by severity, before and after fixes.
3. **Architecture.** Entry points, the top-level Atlas, and the recommended deployment diagram.
4. **Security.** Open findings, fixed findings with verification status, and accepted risks with the builder's written reasons.
5. **Understanding.** The builder can attach path narratives they've reviewed, demonstrating they know how key flows work.
6. **Provenance.** Every tool and version used, every model used, and the rule sets applied.
7. **Limits.** A required, plain-language section stating what the report does not prove, such as runtime behavior not covered by tests, business-logic flaws scanners can't see, and lower accuracy for dynamic code or C++ analyzed without a compile database.

11.2 Verifiability

- Each report gets a public, unguessable link and a PDF export. The builder can revoke the link at any time.
- The report includes a signed digest of its results and the commit hash. A reviewer can re-run the same analysis on that commit with the open-source agent and compare.
- A README badge can show the latest score, linked to the full report.

WHY OPEN-SOURCE THE AGENT

An engineer who doubts the report can read the code that produced it and run it themselves. That is the strongest possible answer to "this is just AI output." The commercial value sits in the hosted experience, the explanations, the PWA, and team features, not in the scanner orchestration.

12 · CLIENT

The PWA client

One installable web app for phone, tablet, and desktop. It is the control panel and the viewer, never the analysis engine.

12.1 Screens

SCREEN	PURPOSE
Home	Repositories, latest score per repo, runs in progress, connected agents and their online status
Atlas · Network	Whole-repo map with module clustering, search, and confidence-styled edges
Atlas · Flow	One entry point's call path as a top-down flowchart; tap a node to open its detail sheet
Node detail	Lens toggle (1 · 2 · 3), source panel, callers and callees, findings on this node, model attribution
Findings	Ranked list with filters; finding detail with lenses, reachability path, diff, and verify status
Readiness	Score breakdown, checklist, advisor output, generated files
Reports	Create, preview, publish, revoke
Settings	Models and keys, routing presets, spend caps, privacy level, connected repos and devices

12.2 Rendering graphs on every screen size

- **Network view** uses WebGL rendering (Sigma.js on a Graphology model) so tens of thousands of nodes stay smooth, with level-of-detail labels.
- **Flow view** uses a layered layout engine (ELK via elkjs) to draw a clean top-down flowchart of one path.
- **On phones**, Flow view defaults to a vertical "path list": each step is a card, indented by call depth, with a swipe to expand callees. The full graph remains available but is never the default on a small screen.
- **Accessibility.** Every graph has an equivalent list representation for screen readers and keyboard navigation. Lens colors are never the only signal.

12.3 PWA capabilities

CAPABILITY	IMPLEMENTATION
Install	Web app manifest with icons and shortcuts; install prompt after the first completed analysis
Offline	Service worker (Workbox) caches the app shell. Graphs, explanations, and reports for recently viewed repos are stored in IndexedDB, so past analyses can be explored on a plane.

CAPABILITY	IMPLEMENTATION
Notifications	Web Push when a run finishes or a new Critical finding appears. On iPhone and iPad, web push requires the PWA to be added to the home screen; onboarding explains this.
Remote control	From the phone, trigger "analyze now" on a paired agent at home, then watch progress stream in over a WebSocket
Local-only use	In Offline privacy level, the agent serves the PWA from localhost, so no data touches Lantern's servers at all

13 · STACK

Technology stack, data model, and API

Choices favor a single portable engine, boring infrastructure, and components the team can replace later.

13.1 Proposed stack

LAYER	CHOICE	RATIONALE
Engine and agent	Rust; tree-sitter grammars compiled in; language servers and scanners run as managed subprocesses	One static binary per platform; fast; memory-safe parsing of untrusted input
Graph format	Versioned schema, serialized as compressed JSON for transport and stored in Postgres with JSONB plus an edges table	Simple to start; graph database only if query needs demand it
Backend API	Python with FastAPI, or TypeScript with Fastify; REST for CRUD, WebSocket for progress	Either is fine; pick the language the founding team ships fastest in
Jobs	Postgres-backed queue initially; move to a dedicated queue when volume requires	Fewer moving parts early
Sandbox	Firecracker microVMs or gVisor on a container platform	Strong isolation for untrusted repos
Storage	Postgres (results, metadata), object storage (reports, PDFs), managed KMS (keys)	Standard and well understood
PWA	SvelteKit or React with Vite; vite-plugin-pwa (Workbox); Sigma.js, Graphology, elkjs	Mature PWA tooling; WebGL graph rendering
Auth	GitHub OAuth plus email magic links; passkeys as an option	Matches the audience; no passwords to leak

13.2 Core entities

ENTITY	KEY FIELDS
User	id, auth identities, default routing preset, spend caps
Repo	id, owner, source (github / local), display name, privacy level, alias map location
Device	id, user, platform, agent version, last seen, token hash, revoked
Run	id, repo, commit, mode, status, stage timings, tool versions, models used, cost

ENTITY	KEY FIELDS
Graph	run, schema version, nodes, edges with confidence, entry points
Finding	id, run, node, tool, rule, SARIF blob, severity, reachability path, status, accepted-risk reason
Explanation	node fingerprint, model, prompt version, lenses, validation result
Report	id, run, public slug, signed digest, revoked

13.3 API sketch

```
POST  /v1/repos                    connect a GitHub repo or register a local one
POST  /v1/repos/{id}/runs          start a run (cloud) or ask a paired agent to run
GET   /v1/runs/{id}                status, stages, cost so far
WS    /v1/runs/{id}/stream         live progress and streamed explanations
GET   /v1/runs/{id}/graph?entry=…  graph, optionally scoped to one entry point
GET   /v1/nodes/{fp}/explanations  all lenses, all models
POST  /v1/nodes/{fp}/explain       generate on demand with a chosen model
GET   /v1/runs/{id}/findings       ranked findings
POST  /v1/findings/{id}/fix        draft a patch; open a PR in cloud mode
POST  /v1/runs/{id}/reports        publish an Evidence Report
POST  /v1/devices/pair             exchange a one-time code for a device token
POST  /v1/agent/results            agent uploads results (signed, chunked)
```

14 · TRUST

Trust and security of Lantern itself

A product that judges other people's security will be judged on its own. This section is a launch gate, not a wish list.

14.1 Threat model highlights

THREAT	MITIGATION
Malicious repository escapes the sandbox	MicroVM isolation; no build execution by default; no credentials inside the VM beyond a short-lived model key; restricted egress
Prompt injection in source code (comments telling the model to report "no issues")	Findings come from scanners, not models, so injection cannot hide a finding. Model output is schema-validated. Source is passed as clearly delimited data. Triage cannot close findings.
Stolen API keys	Envelope encryption; keys decrypted only inside job scope; never logged; local mode keeps keys in the OS keychain

THREAT	MITIGATION
Compromised agent update	Signed releases; reproducible builds; checksums published; update channel pinned to signing key
Stolen device token	Device-scoped tokens, revocable from the PWA; agent requests are signed and rate-limited
Leaked source via reports	Privacy levels; reports default to Structure only; explicit confirmation before publishing snippets
GitHub App over-permission	Read-only contents by default; PR permission requested only on first use; per-repo selection

14.2 Launch gates

- Lantern's own repositories pass Lantern's audit with no open Critical or High findings, and the report is published.
- An independent penetration test of the sandbox and API, with a public summary.
- Published data-handling policy in plain language, written in all three lenses.
- A vulnerability disclosure policy and security contact.

15 · QUALITY

Non-functional requirements

Targets to validate with benchmark repositories in Phase 0. They are design goals, not measured results.

REQUIREMENT	TARGET
Structural analysis, 50k-line Python repo	Under 3 minutes in cloud mode, first run
Incremental re-run after a small change	Under 20 seconds for stages 1–5
First explanation visible	Under 30 seconds after stage 5 completes
Atlas interaction	60 fps pan and zoom up to 20k nodes on a recent laptop; Flow view usable on a mid-range phone
Edge precision (resolved edges)	At least 95% correct on a labeled benchmark per language

REQUIREMENT	TARGET
Lens 3 anchor validity	100% of published pairs resolve to real symbols, lines, or glossary terms (enforced, not sampled)
Availability	99.5% for API and PWA; local mode keeps working when the cloud is down
Agent footprint	Binary under 60 MB, excluding language servers and scanners
Accessibility	WCAG 2.2 AA for all PWA screens

16 · ROADMAP

Roadmap

Each phase ends with something real users can try. Durations are estimates for a small team and will be revised after Phase 0.

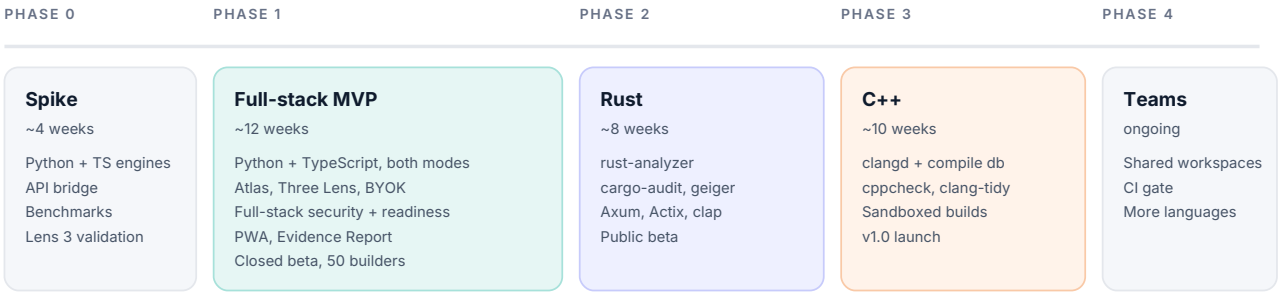


Figure 3. Phased delivery. Phase 1 is deliberately the largest: it builds every layer end to end for the two languages most of the audience ships together, so later languages slot into a finished product.

16.1 Phase 0 exit criteria and status

- **Met (lower bound):** the Python graph's confirmed edges are 95.5–96.7% correct on click, flask, and rich, measured against each project's own test suite at runtime. Hand-labeled samples will tighten the bound.
- **Met:** the full-stack bridge links 23 of 23 calls on the FastAPI template and 5 of 5 on taxonomy, all verified.
- **Open:** Lens 3 generation passes anchor validation on at least 90% of nodes on first attempt with two different model providers. The validator and retry loop are built and tested; real-model runs are next.
- **Open:** five AI-assisted builders and five working engineers react to sample Three-Lens explanations and a mock Evidence Report. Their feedback sets the readiness weights and report layout.

16.2 Business model sketch

TIER	OFFER
Free	Local agent, unlimited local analysis, BYOK or local models, one public Evidence Report
Pro	Cloud mode, private reports, unlimited reports, PR fixes, priority runs
Team	Shared workspaces, CI readiness gate, onboarding maps, admin controls

Explanations start on a free DeepSeek allowance paid by Lantern, capped per user and per day (section 8.5); after that, users pay DeepSeek or another provider directly through their own key. The global daily cap bounds Lantern's model cost.

17 · RISKS AND METRICS

Risks, open questions, and success metrics

The biggest risks are trust risks. Most of the design above exists to reduce them.

17.1 Risks

RISK	IMPACT	MITIGATION
Graphs are wrong for dynamic Python or macro-heavy C++	High	Confidence levels on every edge; runtime tracing; compile databases; honest Limits section
Engineers dismiss the report as AI output	High	Scanner-sourced findings, provenance, open-source agent, reproducible results
Analogies mislead learners	Medium	Lens 3 anchors; consistent metaphor worlds; "this analogy breaks down when..." notes; user feedback per explanation
Model costs surprise users	Medium	Cost preview, spend caps, lazy explanation, aggressive caching
Scanner rule licensing	Medium	Opengrep plus own and permissive rules in cloud; legal review before launch
Sandbox escape from a hostile repo	High	MicroVMs, no build execution by default, pen test as a launch gate
A large incumbent ships something similar	Medium	Focus on the underserved audience; the Mapped lens and Evidence Report as differentiators; community around the open agent

17.2 Open questions

1. Final product name, plus trademark and domain availability.
2. Backend language: Python or TypeScript. Decide by team skills in Phase 0.
3. Should metaphor worlds be chosen by the user, by the model per repository, or from a curated set?
4. Open-source license for the agent: permissive (Apache 2.0) or copyleft (AGPL)?
5. Should the Evidence Report support a reviewer sign-off, letting an engineer attest they reviewed it?
6. Which second code host to add after GitHub, if any, for cloud mode?

17.3 Success metrics

METRIC	WHY IT MATTERS
Median readiness score gain between a repo's first and fifth run	Shows the product actually improves code, not just describes it

METRIC	WHY IT MATTERS
Critical and High findings fixed and verified per active repo	The core safety outcome
Share of users who toggle into Lens 1 after starting in Lens 2	Shows the Mapped lens is working as a learning ramp
Evidence Reports opened by someone other than the author	Measures the product's reach into the engineer audience
Explanation thumbs-down rate, per model and lens	Quality signal for routing defaults and prompt changes
Split of runs between cloud and local mode	Validates the two-mode investment and guides roadmap

RECOMMENDED NEXT STEP

Run Phase 0: a four-week spike on the Python engine and the Lens 3 validator, tested on ten public repositories and reviewed by five builders and five engineers. It answers the two questions that decide everything else: can the graph be trusted, and does the Mapped lens actually teach.

18 · COMPETITION

Competitive landscape

Every piece of Lantern exists somewhere. No product found combines them, and the combination is what persuades a skeptical engineer.

Research as of October 31, 2026. This market moves monthly; re-check before fundraising or launch.

18.1 The four neighboring categories

CATEGORY	EXAMPLES	WHAT THEY DO WELL	WHAT THEY DON'T DO
Deterministic code maps	Graphify (about 140k GitHub stars), repowise, Madge, dependency-cruiser; CodeSee (acquired by GitKraken, 2024; not sold separately); Sourcetrail (archived 2021); AppMap (runtime maps)	Graphs built by parsers. Graphify tags each connection as explicit or inferred and runs as a skill inside AI coding assistants. repowise scores callers by confidence.	No frontend-to-backend linking, no plain-language layer, no security. Built for agents and senior engineers.
AI repository explainers	DeepWiki (Cognition), Google Code Wiki (Gemini), GitDiagram	Fluent, instant documentation for any public repo.	Structure is inferred by a model, so it can be wrong. No anchored plain-English layer, no findings.
Vibe-coding security scanners	Lovable's built-in scanner, Vibe App Scanner, VibeEval, VibeCheck, SafeWeave, amihackable.dev; general tools such as Aikido, Snyk, Semgrep, Supabase Security Advisor	Fast checks for leaked keys, open Supabase or Firebase data, missing headers. Live-site scanners probe the running app from outside.	Lists of problems without the path that triggers them, and little teaching. A reviewer of Lovable's scanner reports that it checks protections exist, not that they work.
AI code review bots	CodeRabbit, Qodo, Greptile	Comments on pull requests.	Diff-focused; no whole-system map or learning path.

18.2 Capability comparison

CAPABILITY	LANTERN	CODE MAPS	AI EXPLAINERS	VIBE SCANNERS
Call graph from parsers, with confidence	Yes	Yes	No	No
Frontend call linked to the backend route	Yes	No	No	No

CAPABILITY	LANTERN	CODE MAPS	AI EXPLAINERS	VIBE SCANNERS
Plain English pinned 1:1 to technical terms	Yes	No	No	No
Security findings	Yes	No	No	Yes
Finding shown with the page and click that trigger it	Yes	No	No	No
Probes the live, deployed app	Not yet (19.1)	No	No	Some
Runs inside AI coding assistants	Not yet (19.1)	Graphify	No	Some
Shareable evidence for a reviewer	Yes: commit-pinned Evidence Report, checkable with `lantern verify`	No	Wiki pages	Score badges

18.3 Where Lantern wins and where it is exposed

ADVANTAGES

Hard to copy quickly

- **One path, button to database.** Needs both language engines plus the bridge; neither map tools nor scanners have the frontend half linked.
- **Findings with the click that triggers them.** Turns "possible SQL injection" into "the profile page reaches this", which is what engineers act on.
- **The Mapped lens.** A teaching asset no one else attempts, enforced by a validator.
- **Every route is known.** Live scanners guess which URLs exist; Lantern has the full list (useful for 19.1).

EXPOSURES

Where competitors are ahead

- **Live verification.** Scanners that read a Supabase table with the public key prove the problem; Lantern currently infers it.
- **Distribution.** Graphify reached its audience as a skill inside Claude Code and Cursor. Lantern has no in-editor presence yet.
- **Platform bundling.** Lovable ships a scanner inside the builder; more builders will follow.
- **Brand of "free and instant".** DeepWiki and Code Wiki need no install. Lantern must match that first-run speed.

POSITIONING

Don't compete as another scanner; that market is crowded and commoditizing. Lantern is the tool that **explains your app and proves it's safe**: one map from click to database, findings shown on that map, plain English pinned to real code, and evidence a reviewer can check.

19 · ROADMAP ADDITIONS

What to add next: standing out, ease, appeal

Ranked by how much each widens the gap with competitors, against its effort. Each maps to a screen in the app design.

19.1 Stand out

FEATURE	WHAT IT DOES	SCREEN	EFFORT
Prove it (live check)	With the user's consent and only against a domain they verify they own, Lantern sends safe test requests to the routes it found: reads Supabase tables with the public key, calls data-changing routes without a login, looks for exposed source maps and missing headers. Findings move from "suspected" to "confirmed", which no static tool can claim and no live scanner can target as precisely.	Finding, Evidence	M
Wallet-drain check for AI apps	Many vibe-coded apps call OpenAI, Anthropic, or DeepSeek from their backend. Flag any route that reaches a paid AI API (or Stripe, Twilio, email) without a login check or rate limit: anyone could run up the owner's bill. Pure graph query; no one else frames it this way.	Finding, Readiness	S
Ask Lantern with a screenshot	The user uploads a screenshot of their own app and asks "what happens when I click this?" A vision model reads the visible text; Lantern matches it to string literals in components to find the page and button, then answers from the graph's real path instead of guessing. (`deepseek-flash` supports images.)	Ask Lantern	M
Fix, then watch it get verified	"Fix with my AI" copies a precise prompt (file, lines, path, fix, test) for Cursor, Claude Code, or Lovable. When the change lands, Lantern re-scans and re-runs tests, and the finding closes only if the scanner agrees.	Fix, Watch tests	M
What changed	Between two commits: new routes, new paths reachable without login, new external services, new secrets exposed to the browser. As a GitHub check, it comments on the pull request.	Evidence, Projects	M
Lantern inside AI assistants	A skill and an MCP server for Claude Code, Cursor, and Codex: the assistant can ask "what calls this route?" and see findings before it writes code. The distribution channel Graphify proved.	(none: integration)	S–M
Data-flow map	Where user data goes: form, route, database table, external service. Answers the questions a privacy review or a first enterprise customer will ask.	Atlas (Advanced)	L

19.2 Easier to use

- **No key to start.** The free DeepSeek allowance (8.5); the first report in under two minutes.
- **Paste a GitHub link.** Cloud mode as the default for first-timers; the local agent for private code.
- **"Fix these 3 first".** Readiness opens with the three highest-impact items in plain words, not a list of fifty. The full list sits behind Advanced.
- **Protections screen.** Show what's already right (login checks found, parameterized queries, row-level security on, CORS restricted) as well as what to add. Beginners need encouragement and engineers want to see coverage. The engine's protections pass already reports "in place", "check", and "add" for each item; the screen should lead with the "in place" ones.
- **Builder integrations.** Detect Lovable, Bolt, v0, and Replit projects synced to GitHub and offer one-click analysis.
- **Editor extension.** Hover a function in VS Code or Cursor to see its three lenses and its path from the UI.

19.3 More appealing

- **Path replay.** An animation of a request travelling from the button through the server to the database: the most shareable thing Lantern can produce, and a natural export for the intro video.
- **Readiness progress.** "From 41 to 87 this week", with the fixes that moved it. Progress keeps people coming back.
- **README badge and evidence link.** A score badge that opens the commit-pinned Evidence Report: social proof for the builder, credibility for the reviewer.
- **Guided tours and "defend your code".** Step-by-step walkthroughs of key paths in any lens, then practice questions a reviewer might ask, generated from the real findings and paths.

19.4 Recommended order

1. Free DeepSeek allowance and the Settings states (done in code; designs in progress).
2. Wallet-drain check: small engine work, big story, feeds the Readiness and Protections screens.
3. Fix and verify loop, matching the app's Finding → Fix → Watch tests → Evidence flow (Evidence and Readiness already exist).
4. Lantern skill and MCP server for AI assistants: the cheapest distribution available. *Built; see chapter 20.*
5. Prove it (live check), with domain verification and consent built first.
6. Ask Lantern with screenshots, What changed, and the data-flow map.

20 · CONNECTOR AND PLUGINS

Lantern as a connector; the three plugin seams

Two directions, kept apart on purpose. Outward: Lantern lives inside the AI assistants its audience already uses. Inward: other people extend Lantern through exactly three seams, and everything else stays closed. Both are implemented in the repository (`lantern mcp`, `lantern packs`, `lantern export`).

20.1 Inside Claude, ChatGPT, Cursor and Claude Code

Claude's connectors and OpenAI's apps in ChatGPT are built on the same open standard, the Model Context Protocol (MCP); Cursor, Claude Code, Windsurf and Codex speak it too. So one Lantern server serves all of them. It runs on the builder's machine over stdio (`claude mcp add lantern -- lantern mcp`; `lantern mcp --setup` prints the JSON for the others). ChatGPT and Claude on the web need a URL, so they reach it through an MCP bridge on a host the builder controls; Lantern never opens a port by itself.

TOOL	WHAT THE ASSISTANT GETS	CHANGES ANYTHING?
`lantern_projects`	The projects Lantern knows, with readiness score and open findings.	No
`lantern_findings`	Findings with file, line, the click path from the page that triggers them, whether outside input reaches the line, and the fix. `id` fetches one finding.	No
`lantern_paths`	What calls a function, what it calls, the confidence of each link, and the path a user takes to reach it. For the assistant to call *before* it edits.	No
`lantern_explain`	The stored three-lens explanation and the source. Never calls a model.	No
`lantern_readiness`	Score, grade, failed checks with fixes, what to do first, protections.	No
`lantern_analyze`	Analyze or re-analyze a folder; reports what was fixed and what is new since last time.	Rewrites the results folder
`lantern_recheck`	After a fix: is this finding gone, still open, or moved? The fix-and-verify loop of 19.1, inside the chat.	Rewrites the results folder
`lantern_verify`	Re-run an Evidence Report's analysis and compare digests, so an assistant can check a report instead of trusting a screenshot.	No

WHAT THE CONNECTOR NEVER GETS

There is no tool to accept, close or dismiss a finding, and asking for one returns a refusal. Accepting a risk needs a person's written reason (`lantern accept` , or the Accept button), and that reason appears in the Evidence Report. So the principle "a model never closes a finding" holds by construction, not by policy. The connector also never runs the project's code: `lantern_verify` refuses a report that was made with a runtime trace, because replaying it would run whatever command the report named.

20.2 "Send to..." sends a reference, not a pasted blob

Every finding card has a **Send to…** button: Claude, ChatGPT, or copy for Cursor and Claude Code. The message names the finding ("Fix Lantern finding 57fa455f7b3b (B602) in the shop project") and tells the assistant to pull the file, line, click path and fix through the connector, to call `lantern_paths` before editing and `lantern_recheck` afterwards, and that it cannot accept or close the finding. With the connector installed, the facts come from Lantern and cannot go stale or be edited in transit. Without it, the message carries one line of essentials, enough to start.

20.3 The badge says exactly what it means

"Verified by Lantern" would read as "this app is secure", which no tool can promise and which engineers will pick at. The badge reads **Lantern · commit 1a2b3c · 0 critical · Oct 31** and links to the Evidence Report, whose Limits section says what was not checked. Its color follows the open findings (critical open, high open, neither), not the readiness score, which stays in the tooltip.

20.4 Connections, not toggles

Settings › Connections lists each assistant by the name it gave in the MCP handshake, with when it last called and which tool: "Claude Code · used 2 minutes ago · last: recheck". A toggle cannot tell whether anything is actually connected; a log can. It stores names and times only, never arguments or code.

20.5 The three plugin seams

SEAM	FORM	WHY IT IS SAFE
Analysis packs	JSON pattern files: security rules (`LP-*` , matched one line at a time) and framework entry points (routes, pages, commands) whose handler names Lantern resolves in its own code map. Built in, in the project (`lantern-packs/`), or the user's. A Django pack ships first; Remix, tRPC, SvelteKit and Rails follow. This is the fastest way to close the framework-coverage gap, and outside contributors can do it.	Not code. Third-party pack findings are always labelled heuristic; pack entry points say which pack registered them and where. Packs only add; nothing in the format can close or re-rank a finding, run code, or read outside the repository.

SEAM	FORM	WHY IT IS SAFE
Explanation providers	Any OpenAI-compatible endpoint chosen in Settings, including local Ollama and LM Studio.	A provider only receives the prompt and returns words; the anchor validator rejects any phrase that does not point at real code, whatever model wrote it.
Exporters	`lantern export --sarif` for GitHub code scanning and SARIF viewers; the JSON results for anything else.	Read-only over the results; accepted risks export as suppressions with their reasons.

Deliberately not seams: scanners that run code, hooks into the pipeline, service connectors that read live infrastructure, and anything that could create, dismiss or re-rank a finding. If those come, they come signed and permission-scoped ("this plugin wants to read your Supabase schema"), after a plugin directory exists. Rule packs first, because they cannot run code.

20.6 Next for the connector

- An HTTP transport with "Sign in with GitHub" (shared with the allowance gateway), so ChatGPT and Claude on the web use Lantern without a bridge.
- A Lantern skill file for Claude Code and Cursor that teaches the assistant the habit: paths before editing, recheck after.
- More packs, with the bench numbers a built-in pack needs before it may claim certainty.